

Study Title: A prospective cohort study to describe the morphology of buboes in patients with bubonic plague using ultrasound imaging

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Confidentiality Statement

This document contains confidential information that must not be disclosed to anyone other than the Sponsor, the Investigator Team, HRA, host organisation, and members of the Research Ethics Committee, unless authorised to do so.

TABLE OF CONTENTS

1.	BACKGROUND AND RATIONALE	9
2.	OBJECTIVES AND OUTCOME MEASURES	10
3.	STUDY DESIGN	10
3.1.	STUDY DESIGN AND STUDY SITES	10
4.	PARTICIPANT IDENTIFICATION.....	10
4.1.	Study Participants	11
4.2.	Inclusion Criteria	11
4.3.	Exclusion Criteria	11
4.4.	Definition of a confirmed case of bubonic plague	11
5.	PROTOCOL PROCEDURES.....	11
5.1.	Recruitment	11
5.2.	Screening and Eligibility Assessment	12
5.3.	Informed Consent	12
5.4.	Enrolment	12
5.5.	Blinding and code-breaking	13
5.6.	Description of study procedures (clinical)	13
5.7.	Sample Handling	14
5.8.	Team training.....	14
5.9.	Early Discontinuation/Withdrawal of Participants	14
5.10.	Definition of End of Study.....	15
6.	SAFETY REPORTING.....	15
7.	STATISTICS AND ANALYSIS	15
7.1.	Statistical Analysis Plan (SAP).....	15
7.2.	Description of the Statistical Methods.....	16
7.3.	Sample Size Determination.....	16
7.4.	Analysis populations	16
7.5.	Procedure for Accounting for Missing, Unused, and Spurious Data.....	16
7.6.	Procedures for Reporting any Deviation(s) from the Original Statistical Plan	16
8.	DATA MANAGEMENT.....	16
8.1.	Source Data.....	16
8.2.	Access to Data.....	17
8.3.	Data Recording and Record Keeping.....	17

9.	QUALITY ASSURANCE PROCEDURES	17
9.1.	Risk assessment	17
9.2.	Study monitoring	18
9.3.	Study Committees.....	18
10.	PROTOCOL DEVIATIONS.....	18
11.	SERIOUS BREACHES	18
12.	EXPECTED RESULTS	19
13.	ETHICAL AND REGULATORY CONSIDERATIONS	19
13.1.	Declaration of Helsinki	19
13.2.	Guidelines for Good Clinical Practice	19
13.3.	Approvals	19
13.4.	Other Ethical Considerations	19
13.5.	Reporting	19
13.6.	Transparency in Research	20
13.7.	Participant Confidentiality	20
13.8.	Expenses and Benefits	20
14.	FINANCE AND INSURANCE	20
14.1.	Funding	20
14.2.	Insurance.....	20
14.3.	Contractual arrangements	20
15.	PUBLICATION POLICY	21
19.	DEVELOPMENT OF A NEW PRODUCT/ PROCESS OR THE GENERATION OF INTELLECTUAL PROPERTY	21
16.	ARCHIVING.....	21
17.	APPENDIX A: AMENDMENT HISTORY.....	22

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SYNOPSIS

Study Title	A prospective cohort study to describe the morphology of buboes in patients with bubonic plague using ultrasound imaging
Study registration	N/A non-interventional study
Sponsor	Institut Pasteur de Madagascar B.P. 1274, Ambatofotsikely – 101 Antananarivo Madagascar
Funder	The MOH Family Foundation
Study Design	Prospective cohort study
Rationale	Bubonic plague is transmitted to humans primarily through the bite of infected fleas. Enlarged lymph nodes - usually axillary, cervical and inguinal - appear in the area closest to the bite. Although humans have lived with the plague for thousands of years, relatively little is known about the evolution and morphology of the bubo. To date, bubo ultrasound has only been performed in one study, a 2018 case report of a patient with bubonic plague in the United States. Ultrasound of enlarged inguinal and suprapubic lymph nodes was performed at two time points and demonstrated increased inguinal lymph node presence on repeat ultrasound. The case report did not describe in detail the ultrasound characteristics of the bubo, nor its evolution, and no further ultrasound imaging studies of buboes have taken place since. Our current understanding of the plague bubo is limited. At present, assessment is restricted to observation and palpation, which have remained unchanged for centuries. This gives us only a very imperfect idea of how the structure of a lymph node and the surrounding tissues change with infection and how they evolve during treatment. The advent of new tools such as low-cost, easily deployable portable ultrasound scanners now gives us the opportunity to fill this gap.
Outcome Measures	Primary: Difference in sonographic characteristics of reactive lymph nodes between patients with bubonic plague and patients with other diseases. Secondary: Accuracy (% difference) of calliper measurements compared to ultrasound measurements.
Study Design	Prospective cohort study with examinations at enrolment and one further visit on 11, with an optional additional visit on day 4.
Study Participants	Patients presenting at participating health centres with enlarged lymph nodes and suspected bubonic plague. Participants will be retrospectively categorised in two groups according to the laboratory test results: 1. Patients with laboratory confirmed or probable bubonic plague 2. Patients who are classified as “non-cases” of plague following laboratory confirmation

Intervention(s)	Ultrasound imaging and calliper measurements of reactive lymph nodes		
Sample Size	The study will use a convenience sampling strategy. The study will enrol as many patients as possible up to 50 patients. Therefore no sample size will be pre-defined for the project		
Expected results	The results would improve knowledge of the structure of a lymph node and surrounding tissues in relation to infection and their evolution during treatment and healing. The results will be shared with the Ministry of Public Health in Madagascar.		
Timetable	Data collection: January 2024 to March 2024 Data analysis: April 2024 to August 2024 Final report writing: August 2024 to October 2024		
Key words	Bubonic plague - Lymph nodes - Ultrasound - Madagascar		
	Objectives	Outcome Measures	Timepoint(s)
Primary	1. Descriptive study of sonographic characteristics of lymph nodes in patients with confirmed or probable bubonic plague 2. Comparison of characteristics with nodes in control group consisting of non-cases of plague	Difference in sonographic characteristics of reactive lymph nodes between patients with confirmed and probable bubonic plague and patients classified as “non-cases” of plague following laboratory confirmation	D1 and D11 (optional additional visit at D4)
Secondary	Comparison of calliper measurements with sonographic measurements	Precision (% difference) of calliper measurements compared to ultrasound measurements	D1 and D11 (optional additional visit at D4)
Intervention(s)	Ultrasound imaging and calliper measurements of reactive lymph nodes		

ABBREVIATIONS

CI	Chief Investigator
CRF	Case Report Form
CTRG	Clinical Trials & Research Governance, University of Oxford
GCP	Good Clinical Practice
GP	General Practitioner
HRA	Health Research Authority
ICF	Informed Consent Form
NHS	National Health Service

RES	Research Ethics Service
OXTREC	Oxford Tropical Research Ethics Committee
PI	Principal Investigator
PIL	Participant/ Patient Information Leaflet
R&D	NHS Trust R&D Department
REC	Research Ethics Committee
SOP	Standard Operating Procedure

1. BACKGROUND AND RATIONALE

Plague is caused by the Gram-negative bacteria, *Yersinia pestis*, and manifests in three main clinical forms with sometimes overlapping clinical syndromes: bubonic plague, septicaemic plague (primary or secondary) and pneumonic plague (primary or secondary). (1) Bubonic plague is the most common clinical form and is characterised by the rapid onset of fever and tender and painful lymphadenopathy, following an incubation period of approximately 2 – 6 days. (2) The Case Fatality Rate (CFR) for bubonic plague is around 20 – 60%. (1, 3)

Bubonic plague is transmitted to humans primarily by the bite of infected fleas (1) and enlarged lymph nodes – typically axillary, cervical and inguinal – usually appear at the area nearest to the bite.

98% of reported plague cases are reported in just two countries (Democratic Republic of Congo (18%) and Madagascar (80%)) but the United States, Peru, China and Mongolia – among others – have also reported cases of plague in recent years. (3) Plague has existed in Madagascar for more than 100 years since 1898. (4) It initially occurred in coastal areas and then spread to the central highlands, and is endemic in Madagascar to this day. (5)

Although humans have lived with plague for thousands of years, relatively little is known about the evolution and morphology of the bubo. To date, ultrasound imaging of the bubo has been conducted in only one study, a 2018 case report of a patient with bubonic plague in the United States. (6) Imaging of enlarged inguinal and suprapubic lymph nodes was conducted at two timepoints and demonstrated increased presence inguinal lymph nodes on repeat imaging. (6) The case report did not discuss the sonographic characteristics of the bubo, nor its evolution in detail, and no further ultrasound imaging studies of buboes have taken place since. Computed tomography of the abdomen was used in a separate case study in 1987. (7)

This study aims to describe the evolution and morphology of bubo and the sonographic characteristics of the bubo compared to enlarged lymph nodes that develop in other diseases. In addition to enhancing our overall understanding of the bubo, the results of this study may contribute to increased use of ultrasound to assess lymph nodes in bubonic plague in clinical and research settings if notable differences are detected between lymph nodes in bubonic plague versus lymph nodes for other diseases such as infectious or inflammatory lymph nodes. This study will observe how the bubo evolves during treatment and remission, or complications associated with plague buboes. At present, our understanding is limited to observation and palpation, which have remained unchanged for centuries. We therefore

have a limited idea of how the structure of a lymph node and the surrounding tissues change with infection and how they evolve during treatment.

Portable ultrasound is a non-invasive, low-cost and easily deployable tool that can shed light on these issues. Depending on the results, there may be implications for monitoring patients during illness and after treatment, as well as for the design of future clinical studies on plague treatment. Alongside this study, local clinicians and research staff will be trained to use ultrasound to conduct lymph node imaging, thus increasing local radiological capacity.

Imaging will be conducted using a Butterfly iQ Pro portable ultrasound device.

2. OBJECTIVES AND OUTCOME MEASURES

Objectives	Outcome Measures	Timepoint(s) of evaluation of this outcome measure (if applicable)
Primary Objective 1. Description of sonographic characteristics of lymph nodes in patients with confirmed and probable bubonic plague 2. Comparison of characteristics with nodes in control group consisting of non-cases of plague	Difference in sonographic characteristics of reactive lymph nodes between patients with confirmed and probable bubonic plague and patients classified as “non-cases” of plague following laboratory confirmation	Ultrasound imaging at D1 and D11 (optional additional visit at D4)
Secondary Objectives Comparison of calliper measurements with sonographic measurements	Precision (% difference) of calliper measurements compared to ultrasound measurements	Ultrasound imaging and calliper measurement at D1 and D11 (optional additional visit at D4)

3. STUDY DESIGN

3.1. Study design and study sites

This is a multi-site prospective cohort study with examinations performed at inclusion and at least one follow-up visit performed at day 11, with an optional additional visit being conducted on day 4 depending on local capacity.

The study will be conducted at health centres in three districts: Ambositra, Manandriana and Ambohimahaso. The health facilities are those participating in the IMASOY (6) study (Open-label, randomised, non-inferiority trial of the efficacy and safety of ciprofloxacin versus aminoglycosides + ciprofloxacin in the treatment of bubonic plague) which received CERBM approval on 10/09/2018 (N° 116 MSANP/CERBM) and OxtREC approval on 24/10/2019 (N° 45-19).

4. PARTICIPANT IDENTIFICATION

4.1. Study Participants

Participants of any age presenting to health facilities.

4.2. Inclusion Criteria

- Suspected bubonic plague AND regional lymphadenopathy
- Willing and able to give informed consent for participation in the study

4.3. Exclusion Criteria

The participant may not enter the study if ANY of the following apply:

- Suppuration to buboes/ lymph nodes
- High pain score making ultrasound imaging intolerable
- Not signing the informed consent form

4.4. Definition of a confirmed case of bubonic plague

Confirmed, probable and non-cases of bubonic plague will be defined according to the WHO plague case definition (see appendix 1). (8)

5. PROTOCOL PROCEDURES

Table 1 – Schedule of procedures

Procedure	D1	D4 (optional)	D11
Screening	X		
Consent	X		
Demographic details	X		
Ultrasound imaging	X	X	X
Clinical assessment and calliper measurement	X	X	X

5.1. Recruitment

The study will be undertaken at health centres participating in the IMASOY trial, as research infrastructure has already been established at these sites, in the districts of Ambositra, Manandriana and Ambohimahaso.

On presentation, participants will be identified and approached by the head clinician and/or *téchnicien d'étude clinique* (TEC; clinical study technician for the IMASOY trial) based at the participating centres.

5.2. Screening and Eligibility Assessment

There will be no exemptions to the eligibility criteria and study procedures must not take place before informed consent has been obtained.

5.3. Informed Consent

The participant must personally sign and date the latest approved version of the Informed Consent form before any study specific procedures are performed.

Written and verbal versions of the Participant Information and Informed Consent will be presented to the participants detailing no less than: the exact nature of the study; what it will involve for the participant; the implications and constraints of the protocol; the known side effects and any risks involved in taking part. It will be clearly stated that the participant is free to withdraw from the study at any time for any reason without prejudice to future care, without affecting their legal rights, and with no obligation to give the reason for withdrawal.

The participant will be allowed as much time as they wish to consider the information, and the opportunity to ask questions of the head clinician of the CSB, the TEC or other independent parties to decide whether they will participate in the study. Written Informed Consent will then be obtained by means of participant-dated signature and dated signature of the person who presented and obtained the Informed Consent. The person who obtained the consent must be suitably qualified and experienced and have been authorised to do so by the Principal Investigator. A copy of the signed Informed Consent will be given to the participant. The original signed form will be retained at the study site.

Patients who are co-enrolled in IMASOY will be asked to consent to the data collected as part of IMASOY to be used in this study.

5.4. Enrolment

This is a non-randomised study.

After signing the Informed Consent form patients will be allocated a participant ID indicating their enrolment to the study. Case selection, consent, inclusion, ultrasound and calliper measurement of the lymph node will take place and demographic data will be collected on the same day.

The first ultrasound examination must take place within 24h of enrolment in the study. The examination can take place either before and/or after bubo aspiration for biological testing (RDT and molecular analysis). The results of these tests will be used to categorise suspected bubonic plague patients.

Examination by ultrasound will be performed on the palpable bubo, and may also be performed at other lymph node sites commonly associated with bubonic plague (axillary, cervical and inguinal) to understand any potential extensive lymph node involvement. These procedures will be performed on days 1 and 11, with the option of performing the assessment on D4 according to local capacity.

Ultrasound examination and calliper measurements will be performed on the same day by two independent observers - one observer will perform the ultrasound and the other will measure the lymph node with a calliper. Each observer will be unaware of the other's measurements to eliminate bias.

Ultrasound data will be recorded using a Butterfly iQ Pro, with a choice of high resolution (musculoskeletal) and lower resolution (abdominal) transducers for in-depth examination of nodal stations. The scanning technique and methodology will be in accordance with the pre-training. All data will be recorded on an electronic case report form (eCRF) to describe bubonic and lymph node characteristics.

Based on laboratory diagnosis, participants will be assigned to one of two groups:

- Patients with a confirmed or probable diagnosis of bubonic plague
- Patients who are classified as “non-cases” of plague following laboratory confirmation

All patients will undergo the same procedures regardless of diagnosis.

5.5. Blinding and code-breaking

This is not a blinded study.

5.6. Description of study procedures (clinical)

Participants will undergo two clinical study procedures:

- Clinical assessment and calliper measurement
- Ultrasound imaging

On Day 1, the patient may undergo these procedures twice – once before the bubo aspiration and once after the bubo aspiration. However, the procedure must only be performed before the bubo aspiration if there is no risk to delaying the start of the patient’s treatment. In this case, the assessment will only take place after aspiration.

Ultrasound imaging

(a) *Device description*: Imaging will be conducted using a Butterfly IQ or Butterfly IQ+ manufactured by Butterfly Network, Inc. (9) Before operating the device, members of the trial team must have undergone a 5-day course on ultrasound imaging for lymph nodes directed by Worldwide Radiology. (10)

(b) *Device safety*: caution should be given where participants have painful buboes/lymph nodes

(c) *Maintenance and storage of device*: follow instructions provided in device software package

Imaging procedure: The risks of undergoing an ultrasound scan are minimal and are limited to discomfort. A trained operator will discuss this with the participant before scanning. The operator will measure the long and the short axis of the bubo or enlarged lymph node at D1 and D11 (and optionally on D4); the short axis is the maximum width of the bubo/enlarged lymph node, where the width (short axis) crosses the long axis perpendicularly as shown in the figure below and described in the relevant specific detailed SOP. Bubo/ lymph node size is calculated by adding both axes. Further assessment of the primary bubo and index lymph node within a cluster will include nodal shape, hilar definition, nodal necrosis, hypervascularity, perinodal reaction, capsule definition and matting to reflect coalescence.

Should a patient have multiple buboes/enlarged lymph nodes, all will be documented for accurate review of lymphadenopathy pattern, although specific measurements and characteristics will be used for primary bubo and index lymph node assessment only.

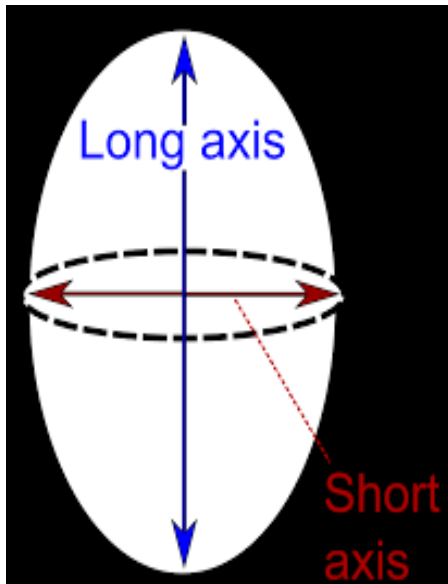


Figure 1a Ultrasound measurements

No samples will be collected as part of this study.

5.8. Team training

An initial intensive ultrasound training session will provide the research team with the practical skills and knowledge necessary to assess lymph nodes by ultrasound from a short, concise, tailored programme. The aim is not to be trained in diagnostic ultrasound which requires longer term dedicated supervision and practical training time.

The research team will receive intensive practical sessions over 5 days in ultrasound examination of lymph nodes. This training session will be accompanied by in-person supervision from a sonographer for 1 week, although remote support will remain in place throughout the data collection period. The limits of this training are well defined, and this training not allow to diagnosis of other clinical problems.

5.9. Early Discontinuation/Withdrawal of Participants

During the course of the study a participant may choose to withdraw early. This may happen for several reasons, including but not limited to:

- The occurrence of what the participant perceives as an intolerable AE
- Inability to comply with study procedures

- Participant decision

There are two scenarios in which patients may wish to withdraw:

- 1) Participants can withdraw from the study but permit data and imaging obtained up until the point of withdrawal to be retained for use in the study analysis. No further data or images would be collected after withdrawal.
- 2) Participants can withdraw completely from the study and withdraw the data and images collected up until the point of withdrawal. The data and images already collected would not be used in the final study analysis.

In addition, the Investigator may discontinue a participant from the study treatment at any time if the Investigator considers it necessary for any reason including, but not limited to:

- Ineligibility (either arising during the study or retrospectively having been overlooked at screening)
- Significant protocol deviation
- Significant non-compliance with the study requirements
- Clinical decision

The type of withdrawal and reason for withdrawal will be recorded on the CRF.

5.10. Definition of End of Study

The end of study is the point at which all the study data has been entered and queries resolved.

6. SAFETY REPORTING

Safety risks relating to the activities of this study are minimal and adverse events are highly unlikely to arise. Therefore, no safety reporting will take place as part of this study.

7. STATISTICS AND ANALYSIS

To describe the characteristics of the bubo, a narrative description comparing the sonographic images of the two participant groups will be provided to meet the primary objectives of the study.

This includes mapping of locations of lymphadenopathy, documentation of the primary bubo and index lymph node characteristics such as shape, cortex homogeneity, hilar and capsular definition, necrosis, vascularity, perinodal reaction and matting for assessment of coalescence and adjacent inflammation.

To understand the relationship between the clinical status of the patient and the evolution of the bubo, bubo characteristics, including size, will also be compared to clinical signs and symptoms.

For the secondary objective, the statistical analysis will assess bubo measurement precision (see **section 11.2**).

7.1. Statistical Analysis Plan (SAP)

The statistical aspects of the study are summarised here with details fully described in a statistical analysis plan that will be finalised before any analysis takes place.

7.2. Description of the Statistical Methods

Descriptive statistics will be used to summarise the differences in 1) morphology and evolution of the bubo and 2) the percentage difference between measurements taken by calliper and by ultrasound.

7.3. Sample Size Determination

As the analyses being undertaken as part of the project is exploratory in nature, no formal sample size will be required. The project will aim to enrol as many participants as possible during the study period.

7.4. Analysis populations

All participants who were enrolled in the study will be included in the descriptive analysis of the ultrasound images.

All participants with confirmed diagnosis of bubonic plague will be included in the measurement precision analysis.

Confirmed, probable and non-cases of bubonic plague will be defined according to the WHO plague case definition (see appendix 1). (8)

7.5. Procedure for Accounting for Missing, Unused, and Spurious Data.

Samples with missing data relating to key variables will be removed from the dataset. See Data Management Plan for further details.

7.6. Procedures for Reporting any Deviation(s) from the Original Statistical Plan

Any deviation(s) from the original analysis plan will be described and justified in the final report.

8. DATA MANAGEMENT

The data management aspects of the study are summarised here with details fully described in the Data Management Plan.

8.1. Source Data

Source documents are where data are first recorded, and from which participants' CRF data are obtained. These include hospital records (from which medical history and previous and concurrent medication may be summarised into the CRF), clinical and office charts, and US images. For patients who are co-enrolled in IMASOY, the clinical data recorded in the IMASOY CRF will be considered source data.

CRF entries will be considered source data if the CRF is the site of the original recording (e.g. there is no other written or electronic record of data). All documents will be stored safely in confidential conditions. On all study-specific documents, other than the signed consent, the participant will be referred to the study participant number/code, not by name.

8.2. Access to Data

Direct access will be granted to authorised representatives from the Sponsor and collaborators for monitoring and/or audit of the study to ensure compliance with regulations.

For patients who are co-enrolled in IMASOY, data from the IMASOY CRF will be merged with the dataset from this study to allow a comparison between the evolution of the patient's clinical status with the evolution of the bubo.

8.3. Data Recording and Record Keeping

Ultrasound images and cine clips will be exported and saved to the Butterfly Cloud and on an external password protected hard-drive while data collection is ongoing. At the end of the study, data will be held on Institut Pasteur de Madagascar servers for 15 years following the end of the study. A copy of the ultrasound images and cineclips will also be held at the Institut Pasteur de Madagascar for this same period.

All other project data will be entered on to an eCRFs database (RedCap) and held on Institut Pasteur de Madagascar servers for 15 years following the end of the study.

The participants will be identified by a unique participant ID allocated at enrolment. The name and any other identifying detail will NOT be included in any trial data electronic file.

9. QUALITY ASSURANCE PROCEDURES

The study may be monitored, or audited in accordance with the current approved protocol, GCP, relevant regulations and standard operating procedures.

9.1. QA for Ultrasound

From an imaging perspective, QA procedures are key to maintain high performance of scan equipment and ensure reliability of results.

Practical and knowledge skills training prior to formal commencing of scanning will ensure standardised methods of scanning, measuring and reviewing.

Equipment QA will cover standardised infection control, assessment for probe damage and basics of display checks.

Imaging will be reviewed by two radiologists following acquisition to ensure ongoing QA, within limitations of patient numbers.

A standardised pathway for communication of urgent incidental findings will ensure patient safety is embedded in the quality assurance process.

9.2. Risk assessment

Minimal risks are associated with the activities included in this study therefore no formal risk assessment will take place.

The usual level of risk associated with the processing of patient data exists for this study as with any other study. The study team will comply with LOI N° 2014 – 038 Sur la protection des données à caractère personnel and organisational policies to minimise the risk of accidental or unlawful destruction, loss, alteration, unauthorised disclosure of, or access to the personal data transferred and stored or otherwise processed by the recipient as part of the data processing involved in this study.

9.3. Study monitoring

Monitoring will be performed on all data and imaging collected as part of this study according to the study specific Monitoring Plan. Data will be evaluated for compliance with the protocol and accuracy in relation to source documents as these are defined in the study specific Monitoring Plan. Following written standard operating procedures, the monitors will verify that the clinical study is conducted and data are generated, documented and reported in compliance with the protocol, GCP and the applicable regulatory requirements.

9.4. Study Committees

A Study Management Group will be established to maintain oversight of the project set-up, data collection and analysis. The Study Management Group will consist of study investigators, those involved in data collection and analysis and radiologists.

A Data Monitoring and Safety Committee is not required due to the short-term nature of the project, minimal safety risks for participants and descriptive nature of the analysis.

10. PROTOCOL DEVIATIONS

A study related deviation is a departure from the ethically approved study protocol or other study document or process (e.g. consent process) or from Good Clinical Practice (GCP) or any applicable regulatory requirements. Any deviations from the protocol will be documented in a protocol deviation form and filed in the study master file.

A standard operating procedure will be in place describing the procedure for identifying non-compliances, escalation to the central team and assessment of whether a non-compliance /deviation may be a potential Serious Breach.

11. SERIOUS BREACHES

A “serious breach” is a breach of the protocol or of the conditions or principles of Good Clinical Practice which is likely to affect to a significant degree –

- (a) the safety or physical or mental integrity of the trial subjects; or
- (b) the scientific value of the research.

In the event that a serious breach is suspected the Sponsor must be contacted within 1 working day. In collaboration with the C.I., the serious breach will be reviewed by the Sponsor and, if appropriate, the Sponsor will report it to the relevant REC within seven calendar days.

12. EXPECTED RESULTS

From a scientific point of view, the study will provide us with more knowledge about how the structure of a lymph node and surrounding tissues change with infection and how they evolve during treatment and recovery. These data will allow us to understand the evolution of the bubo throughout the disease and treatment, and whether the evolution and characteristics of the bubo have any relationship to patient outcomes. The study will also allow for better quality assessment and follow-up of suspected bubonic plague patients.

The results of this study will be shared with the Ministry of Public Health in Madagascar and the study participants.

13. ETHICAL AND REGULATORY CONSIDERATIONS

13.1. Declaration of Helsinki

The Investigator will ensure that this study is conducted in accordance with the principles of the Declaration of Helsinki.

13.2. Guidelines for Good Clinical Practice

The Investigator will ensure that this study is conducted in accordance with relevant regulations and with Good Clinical Practice.

13.3. Approvals

The protocol, informed consent form and participant information sheet will be submitted to Oxford Tropical Research Ethics Committee (OxTREC) and Comité d'Éthique de la Recherche Biomédicale (CERBM) in Madagascar.

The Investigator will submit and, where necessary, obtain approval from the above parties for all substantial amendments to the original approved documents.

13.4. Other Ethical Considerations

In the unlikely event of seeing any additional structural abnormalities on a scan, the scan will be checked by a clinical specialist. If the specialist feels that the abnormality is medically important, they will discuss the implications with the participant's treating clinician who will take appropriate action to investigate and manage the participant's ongoing care. Participants will not be informed unless the doctor considers the finding has clear implications for their current or future health. A nominal remuneration for the cost of investigation and/or treatment (where it is not already freely provided by a national treatment programme) will be offered to participants for whom medically significant incidental findings are detected. In all cases, the results will be explained to the participant. It should be noted that the study scans are not carried out for diagnostic purposes, and therefore the scans are not a substitute for a clinical appointment. Rather, the scans are intended for research purposes only.

13.5. Reporting

The CI shall submit once a year throughout the study, or on request, an Annual Progress report to the REC and Sponsor (where required). In addition, an End of Study notification and final report will be submitted to the same parties.

13.6. Transparency in Research

Not applicable – the research is non-interventional

13.7. Participant Confidentiality

The study will comply with the LOI N° 2014 – 038 Sur la protection des données à caractère personnel, which requires data to be de-identified as soon as it is practical to do so. The processing of the personal data of participants will be minimised by making use of a unique participant study number only on all study documents and any electronic database(s). All documents will be stored securely and only accessible by study staff and authorised personnel. The study staff will safeguard the privacy of participants' personal data.

13.8. Expenses and Benefits

Travel expenses

Participants will be compensated for time lost and travel (equivalent of £6) when attending the follow up visits after the beginning of the treatment. Only those who attend the follow up visit will receive recompense.

Medically significant incidental findings

Participants for whom medically significant incidental findings are identified on ultrasound scans may also receive a contribution towards the cost of their onward care if their care is not covered by a national treatment programme. A nominal remuneration for the cost of investigation and/or treatment (where it is not already freely provided by a national treatment programme) to the equivalent of £65 in the local currency will be offered to participants for whom medically significant incidental findings are detected. (See section 13.4)

14. FINANCE AND INSURANCE

14.1. Funding

The MOH Family Foundation

14.2. Insurance

No insurance is required for this observational study.

14.3. Contractual arrangements

Appropriate contractual arrangements will be put in place with all third parties.

15. PUBLICATION POLICY

The Investigators will be involved in reviewing drafts of the manuscripts, abstracts, press releases and any other publications arising from the study. Authors will acknowledge that the study was funded by the MOH Family Foundation. Authorship will be determined in accordance with the ICMJE guidelines and other contributors will be acknowledged.

19. DEVELOPMENT OF A NEW PRODUCT/ PROCESS OR THE GENERATION OF INTELLECTUAL PROPERTY

Not applicable

16. ARCHIVING

Documents containing the personal information of participants will be securely destroyed following the conclusion of all follow-up visits and review of all study scans.

Essential study documents will be transferred from sites to Institut Pasteur de Madagascar and retained for 15 years following the end of the study.

Research data collected on the eCRF will be downloaded on to encrypted hard drives and securely stored for at least 15 years following the end of the study.

17. APPENDIX A: AMENDMENT HISTORY

Amendment No.	Protocol Version No.	Date issued	Author(s) of changes	Details of main changes made
1	2.0	14Sep2022	J Bourner	• Addition of co-PI • Amendment of the “post-D1” visit to D4 • Amendment of D21 visit to D11 • Addition of section describing ultrasound training • Amendment of data retention period from 5 to 3 years

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Appendix 1 – WHO plague case definition

Plague case – Cas de peste	Suspected – Suspect	Probable – Présumé	Confirmed – Confirmé	Not-a-case – Non cas
Clinical and context – Contexte clinique et épidémiolo- gique	(exposure to infected humans or animals, ET (exposition à des personnes ou des animaux infectés, résidence ou retour d'un foyer endémique connu dans les 10 jours précédant le début de la maladie)	Clinical presentation suggestive of plague AND Epidemiological context suggesting possible exposure to plague (exposure to infected humans or animals, or residence in or travel to a known endemic focus within 10 days prior to onset of the disease) Tableau clinique évocateur de peste ET Contexte épidémiologique évocateur d'exposition	Clinical presentation suggestive of plague AND Epidemiological context suggesting possible exposure to plague (exposure to infected humans or animals, or residence in or travel to a known endemic focus within 10 days prior to onset of the disease) Tableau clinique évocateur de peste ET Contexte épidémiologique évocateur d'exposition	Clinical presentation suggestive of plague AND Epidemiological context suggesting possible exposure to plague (exposure to infected humans or animals, or residence in or travel to a known endemic focus within 10 days prior to onset of the disease) Tableau clinique évocateur de peste ET Contexte épidémiologique évocateur d'exposition
Tests – Tests	NONE – AUCUN	AND ONE of the following: - F1 antigen positive in bubo aspirate, sputum, blood, or post-mortem tissues by F1RDT or DFA - Single anti-F1 serology positive without evidence of previous <i>Y. pestis</i> infection or vaccination - Direct microscopy in a clinical sample, positive for gram-negative coccobacilli that display bipolar staining with Wayson or Giemsa stain ET UN des tests suivants positif: - Détection de l'antigène F1 par TDR F1 ou DFA dans un prélèvement de bubon, de sang, un crachat ou un prélèvement de tissu post-mortem - Une sérologie anti-F1 unique sans signe d'infection antérieure par <i>Y. pestis</i> ni de vaccination - Examen microscopique d'un échantillon clinique mettant en évidence des coccobacilles à Gram négatif, bipolaires après une coloration de Wayson ou de Giemsa	AND at least ONE of the following criteria: - Isolation of <i>Y. pestis</i> from a clinical sample - must have appropriate colony morphology and be identified as <i>Y. pestis</i> based by at least two of the following: - o phage lysis at 20-25°C - o biochemical profile - o F1 antigen detection - Seroconversion or a 4-fold difference in anti-F1 antibody titer in paired serum samples drawn at least 2 weeks apart <i>Y. pestis</i> DNA positive by species-specific PCR on either clinical sample or culture according to standard practice ET au moins UN des tests suivants positif: - Identification de <i>Y. pestis</i> dans un échantillon clinique sur la base de la morphologie de la colonie et au moins DEUX des tests suivants: - o Lyse des cultures à 20-25°C par un phage spécifique - o Profil biochimique de <i>Y. pestis</i> - o Détection de l'antigène F1 - Séroconversion ou multiplication par 4 du titre d'anticorps anti-F1 dans des échantillons de sérum appariés prélevés à au moins 2 semaines d'intervalle - Détection d'ADN de <i>Y. pestis</i> par PCR dans un échantillon clinique ou sur culture, selon la technique conventionnelle	AND either: - at least TWO of the following laboratory tests (F1RDT, DFA against F1 antigen, direct microscopy, convalescent serology, culture, PCR) are conducted AND they are negative OR - When no confirmatory tests can be performed, TWO negative F1RDT on two clinical specimens collected with 24 hours interval ET soit: - Au moins DEUX des tests suivants sont effectués ET sont négatifs : TDR F1, détection de l'antigène F1 par DFA, examen microscopique, sérologie de convalescence, culture, PCR OU - Lorsqu'aucun test de confirmation n'a pu être effectué, DEUX TDR F1 sont négatifs sur deux échantillons cliniques prélevés à 24 heures d'intervalle

DFA: direct fluorescence assay; DNA: deoxyribonucleic acid; F1RDT: rapid diagnostic test based on F1 antigen; PCR: polymerase chain reaction. – DFA: examen par immunofluorescence directe; TDR F: test de diagnostic rapide basé sur l'antigène F1; PCR: de l'anglais «polymerase chain reaction».

World Health Organization = Organisation mondiale de la Santé. (2021). Weekly Epidemiological Record, 2021, vol. 96, 40 [full issue]. Weekly Epidemiological Record = Relevé épidémiologique hebdomadaire, 96 (40), 485 - 496. World Health Organization = Organisation mondiale de la Santé.
<https://apps.who.int/iris/handle/10665/346036>